AERIS: Boundary-Mapped Reliable Routing for Wireless Sensor Networks under Heterogeneous Channels

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***Abstract*—Reliable routing remains a challenging problem in wireless sensor networks under heterogeneous and unstable channel conditions. Existing WSN protocols address this challenge through clustering, chain-based aggregation, threshold-triggered reporting, or collection-tree parent selection. However, classical protocols often suffer from fragile long-range uplinks, while advanced collection-tree designs introduce complex control logic, leaving room for a simple and auditable reliability-first solution. To address these limitations, an adaptive environment-aware routing protocol, AERIS, is proposed. A context-adaptive intra-cluster forwarding mechanism selects direct, chain, or two-hop transmission based on local channel quality, residual energy, and node utilization. A Gateway-assisted uplink mechanism reinforces the fragile cluster-head-to-base-station path. Additionally, a sparse Skeleton reserve backbone serves as a deterministic fallback. Experimental results demonstrate that AERIS ranks first in 21 of 28 evaluation cells and top-two across all 28 cells compared to classical WSN baselines. In a fixed 100-node evaluation, AERIS improves PDR by 30.1, 24.2, 26.0, and 41.4 percentage points over PEGASIS, TEEN, HEED, and LEACH, respectively.**

***Index Terms*—wireless sensor networks, reliable routing, cross-platform evaluation, deployment boundary**

I. Introduction

Wireless sensor network (WSN) routing is traditionally evaluated through the lens of energy efficiency, yet in practical deployments, link stability often dictates the fundamental usability of a routing rule [1]–[3]. Classical protocols address this trade-off via distinct architectural priorities: LEACH focuses on clustered forwarding [4], PEGASIS employs chain-based aggregation [5], HEED incorporates residual-energy-aware head election [6], and TEEN restricts reporting to event-triggered thresholds [7]. In real-world scenarios, the problem is compounded because low-power WSN channels are inherently asymmetric, environment-sensitive, and prone to instability near connectivity thresholds [8]–[10].

The practical challenge is not to identify a single dominant routing family. It is to find the regimes where a simple, auditable controller outperforms classical energy-first designs, and the regimes where richer collection-tree or RPL-style stacks should remain the default. AERIS addresses that question with fixed-rule control rather than learned heuristics. Its architecture combines Context-Adaptive Switching (CAS) for intra-cluster mode selection, Gateway reinforcement for CH-to-BS uplinks, and a sparse Skeleton reserve backbone.

We make three primary contributions:

1) We present AERIS as a fixed-rule routing design whose online decisions are traceable to channel and topology inputs.
2) We evaluate it with a five-protocol NS-3 comparison and an expanded seven-protocol boundary sweep, including simplified CTP/RPL-MRHOF baselines and a collision-aware stress layer, while keeping ranking evidence separate from stress evidence.
3) Through ablation and pooled trade-off statistics, we show that the main reliability gain comes from Gateway support and is accompanied by earlier node death.

II. Related Work

Classical low-overhead WSN protocols continue to define the primary engineering operating points. LEACH and HEED mitigate long-range transmission costs via clustering, though their uplink quality remains highly sensitive to head placement and channel degradation [6], [11]. PEGASIS distributes load through chain forwarding, but this robustness incurs the cost of increased latency and serialization [5]. While TEEN suppresses redundant transmissions, its logic makes performance metrics sensitive to the specific definition of "expected" delivery [7]. LCN work has also examined scalable downward routing for WSN and IoT actuation [12]. Collectively, these protocols and systems represent the reliability-energy-latency trade-off that any contemporary design must confront.

Recent WSN work improves reliability through learning-based routing and optimization [13]–[15]. Context-aware deployment and energy-management studies address a complementary planning axis [16]–[18]. Security-aware and swarm-assisted variants extend the control model under adversarial or malicious settings [19], [20]. Recent surveys summarize the broader move toward AI-driven routing and LLN management [21]–[23], while multi-agent and federated-DDQN designs explore more adaptive controllers [24]–[26]. AERIS shares the context-awareness objective but deliberately keeps runtime logic fixed and auditable.

Collection-tree and RPL-style protocols remain critical because they address the parent-selection problem through structured control planes. CTP constructs trees based on path quality [27], while RPL and MRHOF standardize DAG construction for lossy networks [28], [29]. Opportunistic variants like ORW and ORPL further exploit link diversity [30], [31]. This paper does not seek to replace these mature families. Instead,

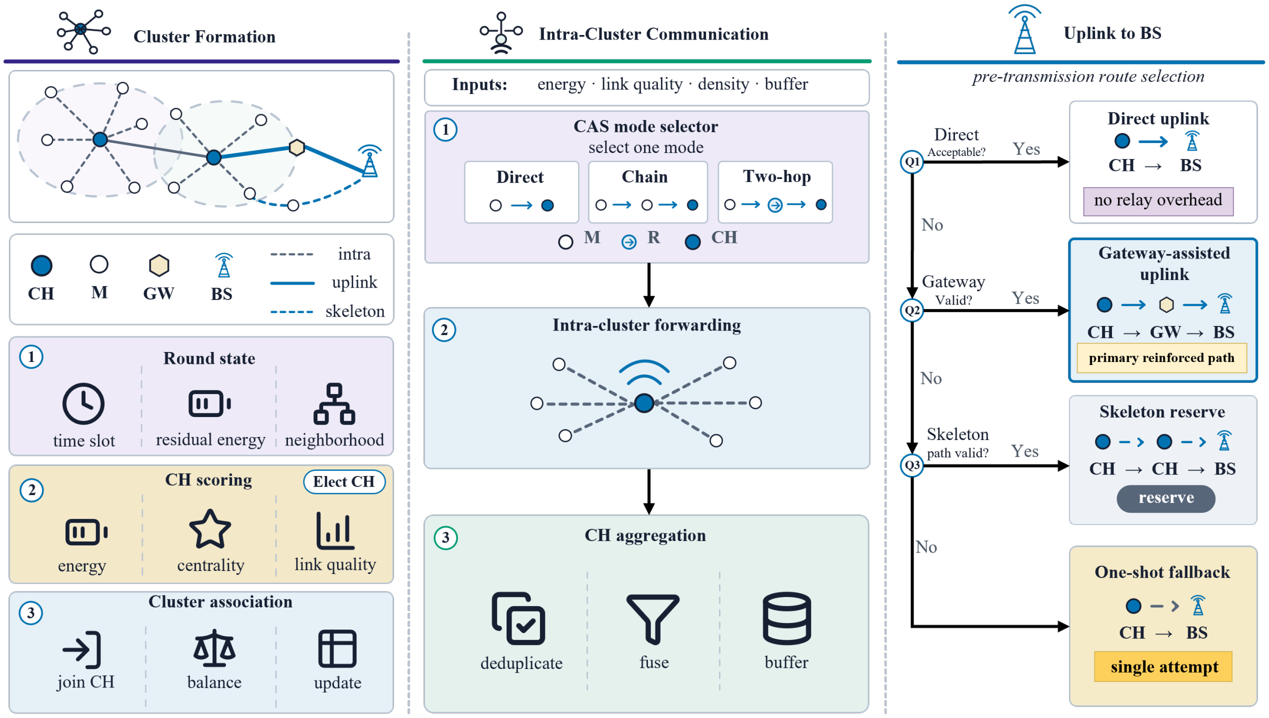

Fig. 1. AERIS round framework. The protocol first forms clusters and scores CH/Gateway candidates, then selects one CAS intra-cluster forwarding mode, and finally applies a first-hit uplink selector over direct, Gateway-assisted, Skeleton-reserve, and one-shot fallback paths.

it investigates where a simpler fixed-rule design provides a sufficient reliability gain and where the established collection-tree paradigm should remain the default choice.

III. Method

This section specifies AERIS as the deterministic round controller in Fig. 1. Its components are cluster formation, context-adaptive intra-cluster forwarding, Gateway-assisted uplink reinforcement, and a sparse Skeleton reserve. The objective is not full collection-tree optimality, but an auditable reliability-first rule set driven by local channel, energy, and topology features.

A. Problem Formulation and Round State

AERIS is a round-based protocol. At the start of each round, every live node exposes a compact local state vector

$$\mathbf{x}_i = [\hat{E}_i, \hat{L}_i, \hat{D}_{iB}, \hat{C}_i, \hat{\rho}_i, \hat{U}_i, p_i]^\top. \tag{1}$$

Here $\hat{E}_i$ is normalized residual energy, $\hat{L}_i$ is the current link-quality estimate, $\hat{D}_{iB}$ is normalized distance to the base station (BS), $\hat{C}_i$ is a neighborhood-centrality surrogate, $\hat{\rho}_i$ represents local density, $\hat{U}_i$ denotes recent forwarding utilization, and p_i is the historical cluster-head (CH) duty penalty. All variables are normalized to $[0, 1]$. A live source contributes one expected sensing report per round; delivery requires both member-to-CH and CH-to-BS success, so the objective is complete collection delivery rather than hop-local reception.

B. AERIS Framework

The control flow in round r is

$$\mathcal{R}_r : \; \mathsf{refresh} \rightarrow \mathsf{CH/join} \rightarrow \mathsf{CAS} \rightarrow \mathsf{uplink} \rightarrow \mathsf{update}. \tag{2}$$

After state refresh, CHs are elected, ordinary nodes associate with CHs, CAS chooses one local forwarding mode, and CH traffic is forwarded through a first-hit uplink selector: direct, Gateway, Skeleton, then one-shot fallback. The selector stops after the first eligible path succeeds; unused fallback stages consume no energy and are not counted as active mechanisms.

C. Cluster Formation

CH election uses a normalized suitability score over residual energy, node centrality, neighborhood degree, distance to the BS, and link quality:

$$p_i^{\mathrm{CH}} = \mathcal{F}(\hat{E}_i, \hat{C}_i, \hat{d}_i, \hat{D}_{iB}, \hat{L}_i)\,(1 - 0.25\,p_i), \tag{3}$$

where $\mathcal{F}$ is the fuzzy CH score used by the implementation, and $\hat{d}_i$ is the normalized degree surrogate. A fairness penalty reduces repeated CH duty through the factor $1 - 0.25p_i$, where p_i records prior CH use. This makes CH election energy-aware without turning CH rotation into the dominant objective.

After CH election, non-CH nodes join one CH using a fixed weighted rule over distance, link quality, and residual energy:

$$J_{ij}^{\mathrm{join}} = 0.45\,(1 - \bar{d}_{ij}) + 0.40\,\hat{L}_{ij} + 0.15\,\hat{E}_j. \tag{4}$$

The node then chooses $j^\star = \arg\max_{j \in \mathcal{F}_i} J_{ij}^{\mathrm{join}}$. The exact weights and feasibility threshold are reported with the experimental configuration in Table II. If no feasible CH is available, the node attaches to the best available CH and the later uplink selector handles the weak path. This separation keeps local cluster association simple while leaving long-range reliability decisions to the Gateway/Skeleton layer.

D. Gateway-Assisted Uplink

The Gateway layer targets the fragile CH-to-BS segment. A CH that is close to the BS or already has a reliable uplink sends directly. Otherwise, AERIS searches for a better-positioned CH that can serve as an intermediate relay. Gateway candidates are ranked by a fixed linear score,

$$S_i^{\mathrm{GW}} = w_D \bar{D}_i + w_C \hat{C}_i + w_L \hat{L}_i + w_E \hat{E}_i. \tag{5}$$

where $\bar{D}_i$ is normalized CH-to-BS distance, and $\hat{C}_i$, $\hat{L}_i$, and $\hat{E}_i$ are normalized centrality, link quality, and residual energy. The highest-score candidate $g^\star = \arg\max_{g \in \mathcal{G}_k} S_g^{\mathrm{GW}}$ is selected, and the fixed publication weights are reported with the experimental configuration in Section IV-A.

The selected Gateway aggregates the originating CH's payload with its own CH traffic and forwards the resulting packet toward the BS. If the Gateway uplink fails, AERIS permits a bounded direct rescue from the originating CH. This behavior is deliberately modest: Gateway support is not a full collection tree and does not maintain a long-lived parent structure. It is a local reliability reinforcement mechanism for rounds in which the direct CH-to-BS path is weak.

Gateway activation is controlled by eligibility rather than by a global route recomputation. For a source CH k, the candidate set is

$$\mathcal{G}_k = \{g \in \mathcal{H} : E_g > 0,\ D_{gB} < D_{kB},\ q_{kg} q_{gB} \geq q_{kB}\}, \tag{6}$$

where $\mathcal{H}$ is the live CH set and q_{ab} is the estimated one-hop success probability between nodes a and b. This filter requires a candidate to be live, geometrically useful relative to the BS, and not weaker than the direct uplink it replaces. Among eligible candidates, the highest fixed score is chosen. If no candidate satisfies the eligibility filter, the CH does not search deeper; it either uses the Skeleton reserve if available or falls back to direct transmission. This shallow decision tree is important for the paper's positioning. It explains why AERIS can be described as auditable, but it also explains why richer CTP/RPL-style baselines can dominate in regimes where repeated parent repair and path-cost updates are beneficial.

E. Context-Adaptive Intra-Cluster Forwarding

CAS chooses how member traffic reaches the CH before the CH attempts its uplink. It compares direct, chain, and two-hop forwarding using a fixed linear mode score:

$$m_c^\star = \arg\max_{m \in \mathcal{M}} \theta_m^\top \phi_c, \quad \mathcal{M} = \{D, C, H_2\}. \tag{7}$$

Here D, C, and H_2 denote direct, chain, and two-hop forwarding. The vector ϕ_c stacks the cluster-level energy, link quality, BS distance, radius, density, and fairness features. Rule overrides are applied only for obvious reliability cases, such as weak long local links where a short two-hop detour remains available.

The three CAS modes have distinct roles. Direct mode minimizes forwarding overhead when intra-cluster links are already reliable. Chain mode serializes local traffic through short hops in denser clusters, which can reduce weak long local links but may increase latency. Two-hop mode inserts one short relay between a member and its CH, which is most useful when local links are unstable but a nearby relay remains viable. CAS therefore does not claim to be a universally beneficial module; it is an explicit switch that exposes when local detours are selected and when the protocol falls back to simpler forwarding.

F. Skeleton Reserve Path

The Skeleton layer is a deterministic reserve, not the primary gain carrier. A sparse set of CHs is selected along a principal spatial axis to provide a fallback relay backbone when direct and Gateway paths are not usable. The reserve path is intentionally conservative: it is activated only after earlier uplink choices fail eligibility or delivery, and it does not continuously maintain a dense mesh. This choice limits control overhead but also explains the mechanism results in Section IV: in the audited configuration, Gateway support resolves most eligible fallback cases, so Skeleton usage remains mostly dormant.

The resulting uplink decision is a first-hit cascade rather than a multipath search:

$$u_k^\star = \begin{cases} \text{direct}, & \chi_k^{\mathrm{dir}} = 1, \\ \text{GW}, & \chi_k^{\mathrm{GW}} = 1, \\ \text{Sk}, & \chi_k^{\mathrm{Sk}} = 1, \\ \text{fallback}, & \text{otherwise}. \end{cases} \tag{8}$$

G. Failure Handling and State Update

AERIS treats a forwarding attempt as failed when the selected hop does not deliver under the current channel realization after the configured attempt budget. A direct CH-to-BS success ends the uplink stage. A Gateway success is counted only after both the source-CH-to-Gateway hop and the Gateway-to-BS hop succeed. Skeleton success is counted only if the reserve path reaches the BS. These definitions are used consistently in the NS-3 and stress-layer accounting so that the reported end-to-end PDR measures complete delivery rather than partial hop success.

For source i in round r, end-to-end delivery is recorded as

$$d_i^{(r)} = \mathbf{1}\{z_{i,c(i)}^{(r)} = 1\}\mathbf{1}\{z_{c(i),B}^{(r)} = 1\}, \tag{9}$$

where $c(i)$ is the assigned CH, $z_{i,c(i)}^{(r)}$ records successful intra-cluster delivery, and $z_{c(i),B}^{(r)}$ records successful CH-to-BS delivery through the selected uplink stage. This definition is strict: a packet that reaches an intermediate relay but not the BS is not counted as delivered.

After each round, residual energy is decremented for transmission and reception events,

$$E_i^{(r+1)} = \max\{0, E_i^{(r)} - \epsilon_{\mathrm{tx}} T_i^{(r)} - \epsilon_{\mathrm{rx}} R_i^{(r)}\}, \tag{10}$$

where $T_i^{(r)}$ and $R_i^{(r)}$ are the numbers of transmit and receive events charged to node i in round r. Link summaries are then refreshed and CH-duty penalties are updated. A node that exhausts its energy is removed from the candidate pool in subsequent rounds. This update rule is the source of the reliability-lifetime trade-off observed later: the Gateway path can protect delivery in harsh cells, but the same relays may accumulate forwarding load and trigger early first-node death. AERIS therefore does not hide relay concentration; it exposes the concentration through the mechanism and lifetime metrics.

H. Packet and Aggregation Semantics

The protocol is evaluated as a data-collection workload in which each live source contributes one expected application reading $s_i^{(r)}$ per round. For cluster k, the CH payload is

$$A_k^{(r)} = \mathcal{A}\{s_i^{(r)} : c(i) = k,\ z_{i,k}^{(r)} = 1\}, \tag{11}$$

where $\mathcal{A}$ is the aggregation operator and only readings that reach the CH enter the payload. A successful end-to-end delivery requires that $A_k^{(r)}$ reach the BS through one uplink path. If a source-to-Gateway hop succeeds but the Gateway-to-BS hop fails, the source readings are not counted as delivered. This strict accounting is why the paper reports PDR over expected application-layer transmissions rather than hop-local reception.

The same semantics also make TEEN comparisons delicate. TEEN is event-triggered and may intentionally suppress transmissions. The evaluation uses a fixed expected denominator so that suppression is not mistaken for reliable delivery. This choice is appropriate for a reliability-first collection workload, but it is not the only possible interpretation of TEEN. A TEEN-centered study should also report event-detection success or actual-transmission PDR. In this paper, TEEN is included to represent a classical threshold-triggered WSN baseline, and its results are interpreted together with energy, lifetime, FND, and hop count.

I. Information Boundary

AERIS is restricted to round-local information. For node i in round r,

$$\mathcal{I}_i^{(r)} = \{\mathbf{x}_i^{(r)}, \mathbf{x}_{\mathcal{N}_i}^{(r-1)}, c(i), \mathcal{H}^{(r)}, \mathcal{G}_{c(i)}^{(r)}, \mathcal{S}^{(r)}\}, \tag{12}$$

where $\mathcal{N}_i$ is the local neighborhood, $c(i)$ is the assigned CH, $\mathcal{H}$ is the CH set, $\mathcal{G}_{c(i)}$ is the Gateway candidate set for that CH, and $\mathcal{S}$ is the reserve Skeleton set. The executed action is

$$a_i^{(r)} = f_\Theta(\mathcal{I}_i^{(r)}), \qquad \Theta \text{ fixed for all } (e, n). \tag{13}$$

Thus AERIS does not use a mobile sink, global topology map, synchronized channel hopping, time-slotted scheduling, DODAG repair, or Trickle dissemination. If no eligible Gateway or Skeleton path exists, the selector returns to direct rescue rather than inventing an unmodeled route.

J. Control Cost

Let N be the number of live nodes, K the number of CHs, and $|\mathcal{C}_k|$ the size of cluster k. The standalone implementation has

$$T_{\text{round}} = O(N)_{\text{CH}} + O(NK)_{\text{join}} + O(K^2)_{\text{GW}} + O(K)_{\text{CAS}}, \tag{14}$$

and local storage

$$M_i = O(1), \qquad M_k^{\text{CH}} = O(|\mathcal{C}_k| + K). \tag{15}$$

No training, route-table diffusion, DODAG repair, or global path optimization is performed during the reported runs. These bounds specify the implementation envelope rather than an optimality claim: AERIS trades the richer repair behavior of collection-tree stacks for a smaller and directly inspectable control path.

K. Decision Precedence

CAS and Gateway operate on different stages. CAS chooses member-to-CH delivery; Gateway and Skeleton operate only after the CH has an aggregated uplink payload. The uplink precedence is

$$\begin{aligned} \Pi_k &= (\text{dir}, \text{GW}, \text{Sk}, \text{fb}), \\ u_k^\star &= \min_{\ell \in \Pi_k} \{\ell : \chi_{k,\ell} = 1,\ z_{k,\ell} = 1\}. \end{aligned} \tag{16}$$

For the two multi-hop uplink stages,

$$z_{k,\text{GW}} = z_{k,g^\star} z_{g^\star,B}, \qquad z_{k,\text{Sk}} = \prod_{(a,b) \in P_k^{\text{Sk}}} z_{ab}. \tag{17}$$

A failed stage advances only this local selector; it does not flood route discovery or build a new tree. Successful Gateways are not persistent parents in later rounds. This first-hit rule also explains the mechanism results: a dormant Skeleton count means earlier Gateway decisions resolved most eligible fallback cases, not that the reserve rule is absent.

L. Collection-Tree Boundary

AERIS and collection-tree routing both address weak multi-hop delivery, but with different state assumptions. CTP/RPL-style designs maintain parent/path-cost state and can repair routes through a richer control plane. AERIS instead uses one-hop CH-level reinforcement and a sparse reserve path:

$$\mathcal{P}_{\text{AERIS}} \subseteq \{k \to B,\ k \to g \to B,\ P_k^{\text{Sk}},\ k \to B_{\text{fb}}\}, \tag{18}$$

while RPL/CTP-style baselines search a larger parent/path space. The evaluation therefore separates the classical-baseline audit from the expanded seven-protocol boundary sweep. The expected result is not universal dominance; if stable parent selection already identifies good paths, a richer collection-tree baseline can remove AERIS's advantage.

IV. Experiments

A. Experimental Setup

The primary metric is packet delivery ratio over expected application-layer transmissions,

$$\text{PDR}_{\text{expected}} = \frac{N_{\text{delivered}}}{N_{\text{expected}}}, \tag{19}$$

where N_{expected} counts one packet per alive source node per round even if a protocol suppresses actual transmission. The denominator is kept fixed across protocols so that the reliability comparison is not confounded by protocol-specific reporting logic.

For Fig. 2, the plotted value is the signed classical margin

$$\Delta_{e,n} = \mathrm{PDR}_{\mathrm{AERIS}}(e,n) - \max_{b\in\mathcal{B}_{\mathrm{classical}}} \mathrm{PDR}_b(e,n), \quad (20)$$

where $\mathcal{B}_{\mathrm{classical}} = \{\mathrm{LEACH}, \mathrm{PEGASIS}, \mathrm{HEED}, \mathrm{TEEN}\}$. Positive values mean that AERIS outperforms the strongest classical baseline in the corresponding environment-node cell.

We use four complementary settings. A five-protocol NS-3 comparison evaluates AERIS against LEACH, PEGASIS, HEED, and TEEN across four environments and seven node scales from 50 to 1000 nodes ($n = 30$ per cell). An expanded seven-protocol boundary sweep adds simplified CTP- and RPL-MRHOF-style collection baselines over the same scales. In this standalone harness, CTP and RPL-MRHOF are simplified collection-style baselines rather than complete LLN stacks. Our CTP variant follows the link-cost forwarding rule of [27] but omits the 4-bit beacon-driven link estimator, dynamic beacon timer, and datapath validation. Our RPL-MRHOF variant follows the path-cost parent selection of [28], [29] but omits trickle-driven DIO/DAO, full DODAG repair, and rank-based loop avoidance. These simplifications remove parts of the standards-compliant control plane and repair machinery, so the AERIS–baseline results should be read as a controlled expanded seven-protocol boundary sweep. Full Contiki-NG/TinyOS stacks may shift the boundary in either direction. Opportunistic variants such as ORW/ORPL [30], [31] and TSCH-based stacks [32] are deferred to future work.

The baselines are therefore not used for a single leaderboard claim. LEACH, PEGASIS, HEED, and TEEN define the classical WSN comparison. The CTP- and RPL-MRHOF-style variants define a harder boundary test by introducing stronger parent/path selection. The stress layer is kept separate because the classical protocols are minimally relay-enabled there for multi-hop comparability. This separation prevents the paper from mixing canonical baseline ranking, adapted stress behavior, and mechanism diagnosis.

A 3360-run NS-3 ablation compares full AERIS with no-Gateway, no-CAS, and no-CH-score variants. A collision-aware Python stress layer adds explicit collision and relay contention; LEACH, HEED, and TEEN are minimally relay-enabled there for multi-hop comparability. A separate fixed 100-node publication block reports PDR, consumed energy, lifetime, first-node death, and hop count pooled across the four environments. A 400-replicate-per-cell mechanism study covers 12 cells (four environments at 100, 500, and 1000 nodes) and explains which AERIS mechanisms actually fire.

The NS-3 evidence uses a standalone ns-3.40 validation harness. Nodes are uniformly placed in a 200 m $\times$ 200 m field with the sink at the top-center boundary. Runs use 300 rounds, 512-byte packets, 2.0 J initial energy, 10 dBm transmit power, and seeds 42001–42030. The channel combines log-distance loss, shadowing, fading, receiver sensitivity, and packet-error probability. The path-loss/shadowing pairs are office (2.0, 4.5 dB), factory (2.7, 8.5 dB), suburban (2.8, 7.5 dB), and urban (3.4, 12.0 dB). The harness operates above an idealized contention-free MAC abstraction rather than the full IEEE 802.15.4 stack. The collision-aware Python stress layer adds explicit collision and relay contention on top of this abstraction.

TABLE I
NS-3 HARNESS PARAMETERS USED FOR THE CANONICAL AUDIT AND EXPANDED SEVEN-PROTOCOL BOUNDARY SWEEP.

Parameter	Setting
Simulator	standalone ns-3.40 harness
Deployment area	200 m $\times$ 200 m, uniform nodes
Sink position	top-center boundary
Node scales (audit/sweep)	50, 100, 200, 300, 500, 800, 1000
Rounds per run	300
Replicates	30 seeds per environment-scale-protocol cell
Seed block	42001–42030
Packet size	512 bytes
Initial energy	2.0 J per node
Transmit power	10 dBm
Environment model	log-distance loss + shadowing + fading
Office/factory	$(n, \sigma) = (2.0, 4.5)$, $(2.7, 8.5)$ dB
Suburban/urban	$(n, \sigma) = (2.8, 7.5)$, $(3.4, 12.0)$ dB
MAC abstraction	idealized contention-free canonical layer

TABLE II
FIXED AERIS CONFIGURATION USED IN ALL REPORTED EXPERIMENTS.

Item	Setting
CH fraction	starts at 0.08; clipped to $[0.05, 0.20]$
CH score	fuzzy chance with fallback weights $0.4/0.3/0.2/0.1$
Join score	distance/link/energy = $0.45/0.40/0.15$
Minimum join PRR	0.20 when a feasible CH exists
Gateway score	$w_D/w_C/w_L/w_E = -0.7/0.3/0/0$
Gateway recovery	one relay retry plus one direct rescue
Skeleton	sparse PCA-axis reserve fallback
CAS smoothing	EMA/confidence = $0.2/0.2$
CAS direct row	$(.35, .65, -.25, -.05, .10, -.05)$
CAS chain row	$(.30, .40, .20, .20, .20, -.05)$
CAS two-hop row	$(.20, .25, .50, .15, .05, -.05)$

All AERIS coefficients are fixed once before evaluation. Table II lists the publication configuration; no coefficient is retuned between office, factory, suburban, and urban cells. The ablation disables one component at a time while leaving the remaining settings unchanged, so it measures module contribution under the same protocol configuration rather than a retuned variant.

The statistical unit is an environment-scale-protocol cell. Figure 2 plots the five-protocol canonical audit as AERIS margin over the strongest classical baseline. This keeps the main classical result readable without line overlap and avoids pretending that the canonical audit is a boundary plot. Against classical WSN protocols alone, AERIS is rank-1 in 21 of 28 environment-node cells and top-2 in all 28, which is enough to justify the claim that it is a reliability-first alternative to the classical baselines.

The expanded seven-protocol boundary sweep is summarized in text rather than as a separate figure. Once simplified CTP- and RPL-MRHOF-style baselines are included, AERIS remains strongest mainly in outdoor suburban cells; office and most factory/urban cells are better covered by the collection-tree baselines. We therefore treat the expanded seven-protocol boundary sweep as deployment-boundary evidence, not as a replacement for the classical audit.

The ablation uses the same seed structure and environment-

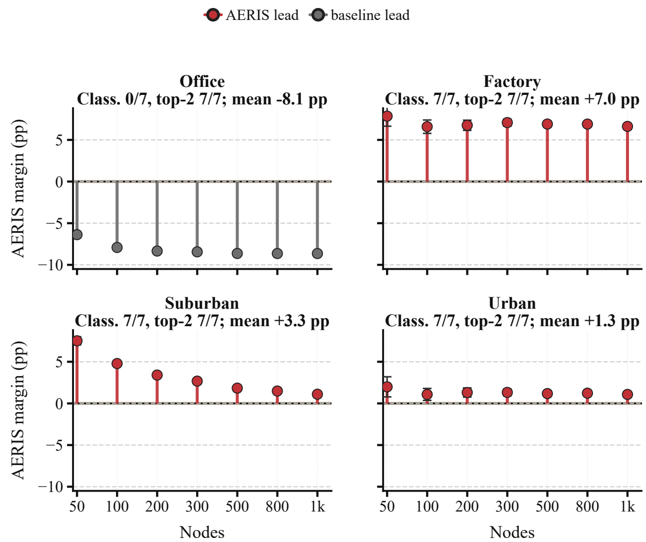

Fig. 2. Canonical NS-3 audit across four environments and 50–1000 nodes ($n = 30$ per cell). Each marker reports AERIS minus the strongest classical baseline in percentage points; whiskers show approximate 95% confidence intervals. Red markers favor AERIS, dark gray markers favor the strongest classical baseline, and the light band marks the ± 0.1 percentage-point near-tie region. "Class." in each panel title reports AERIS rank-1 coverage against LEACH, PEGASIS, HEED, and TEEN; "top-2" reports its top-two coverage in the same classical audit. Marker magnitudes encode the actual percentage-point margin, so small positive markers still indicate AERIS rank-1 cells.

scale grid, but compares full AERIS with one disabled component at a time. Figure 4 reports full-minus-ablated PDR deltas in percentage points; positive values mean that removing the module hurts delivery, while negative values mean that the disabled variant delivered more packets in that cell. Statistical support is computed per environment across the seven node scales after Holm correction. This convention is deliberately conservative: it avoids using the 3360-run count as if every packet or round were an independent statistical sample.

The five-protocol NS-3 comparison anchors the classical-baseline ranking; the expanded seven-protocol boundary sweep provides the deployment boundary; the ablation attributes the gain; and the collision-aware stress layer provides stress evidence rather than a second canonical ranking. Lifetime, FND, hops, and energy expose load concentration. For Fig. 4, module effects are checked with Welch-type cell tests, Hedges' g, and Holm correction. In Fig. 2, winner labels should not be over-read because the plot summarizes mean margins rather than a significance map.

B. Classical Audit and Boundary Note

Figure 2 gives the canonical five-protocol result as signed PDR margins against the strongest classical baseline. This replaces a protocol-line plot because many high-PDR curves overlap visually near the top of the axis; the signed margin exposes the actual classical gain more directly. PEGASIS remains best in indoor office, while AERIS leads over the strongest classical baseline in factory, suburban, and urban conditions. The office gap is negative by design, because PEGASIS is the strongest benign-regime baseline in this evidence package. PEGASIS-only gaps are larger in some harsh cells, but Fig. 2 is the conservative basis for the classical claim because it compares against the strongest classical baseline in each cell. The classical result therefore supports a bounded claim: AERIS is a reliability-first design that remains effective outside the benign office regime.

The expanded seven-protocol boundary sweep, which is summarized in text rather than plotted, shows why that claim must still be bounded. Once simplified CTP- and RPL-MRHOF-style baselines are included, AERIS remains strongest mainly in outdoor suburban cells: it wins 6 of 7 scales and is essentially tied at 1000 nodes, where the simplified RPL-MRHOF-style baseline leads by only 0.01 percentage points. In contrast, the RPL-MRHOF-style baseline leads the office cells and most factory and urban cells. AERIS is slightly ahead at factory-50, but falls 1.3–1.8 percentage points behind the RPL-MRHOF-style baseline at larger factory scales; in urban, the average gap to the winner is 5.69 points. AERIS mitigates the fragility of clustering and chain routing by protecting uplinks with Gateway support, but collection-tree and RPL-style baselines already cover part of the same path-selection problem with richer parent choice.

The strongest defensible claim is therefore not that AERIS is globally best, but that its rule-based control path delivers a large gain over classical WSN baselines and remains competitive or best in selected harsh-channel regimes, especially suburban cells.

C. Collision-Aware Stress Evidence

The collision-aware stress layer asks whether the classical-baseline ordering survives once collision and relay contention are made explicit. Figure 3 suggests that the ordering is largely preserved under this adapted stress model. PEGASIS remains strongest in indoor office (0.991 at 100 nodes and 0.988 at 1000), whereas AERIS stays rank-1 in indoor factory, outdoor suburban, and outdoor urban. At 1000 nodes, AERIS remains well above PEGASIS in indoor factory (0.728 vs. 0.300) and outdoor suburban (0.728 vs. 0.473), and it still leads in outdoor urban (0.136 vs. 0.099).

Because LEACH, HEED, and TEEN are minimally augmented with a simple relay layer for multi-hop comparability in this setting, Fig. 3 is stress evidence rather than a second canonical ranking. Its narrower role is to show that AERIS's harsh-channel advantage survives explicit collision-and-relay stress, even though absolute PDR falls for every protocol in the harder scales.

D. Ablation

Figure 4 compresses the full 50–1000 node sweep into a single-column heatmap instead of a wide line plot. Each panel is one disabled module; color encodes full-minus-ablated PDR in percentage points, and filled dots mark Holm-significant cells. This compact view keeps all seven node scales visible without line overlap, but it should be read as a compressed module-effect summary rather than a direct point-by-point trend plot. Gateway is still the main gain carrier in factory and

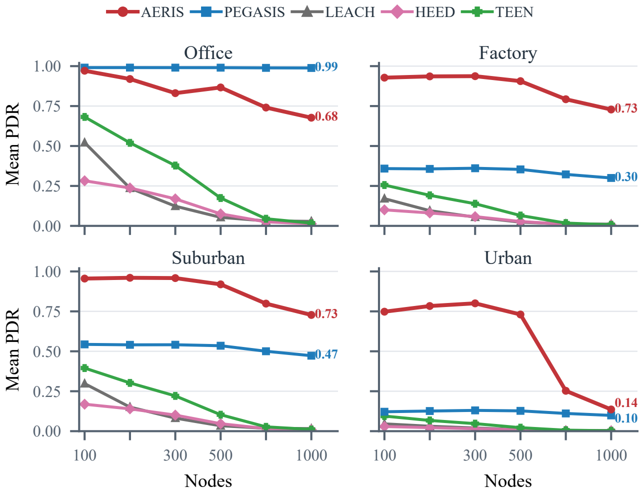

Fig. 3. Collision-aware adapted Python stress test across four environments and 100–1000 nodes ($n = 3200$ per cell). LEACH, HEED, and TEEN are minimally relay-enabled; this layer tests collision-and-relay stress and is not canonical ranking evidence.

suburban cells across the full grid. Removing Gateway costs 5.82 percentage points on average in factory and 7.58 points in suburban, with all seven node scales significant after Holm correction in both environments. It is essentially neutral in office and weak in urban, which matches the boundary result: office does not need extra uplink protection, while urban links can become too poor for a nearby Gateway to rescue reliably.

CAS has a more mixed role. Removing CAS is beneficial in office and suburban cells, but hurts urban delivery by 0.91 points on average, with five of seven urban scales significant. CAS is therefore not a universal gain module; it is most useful when a short local detour remains viable under harsh links. Removing the CH scoring component is near-neutral in all environments. That does not make the score irrelevant to the design, but it does show that the measured delivery gain in this publication configuration should not be attributed to head scoring.

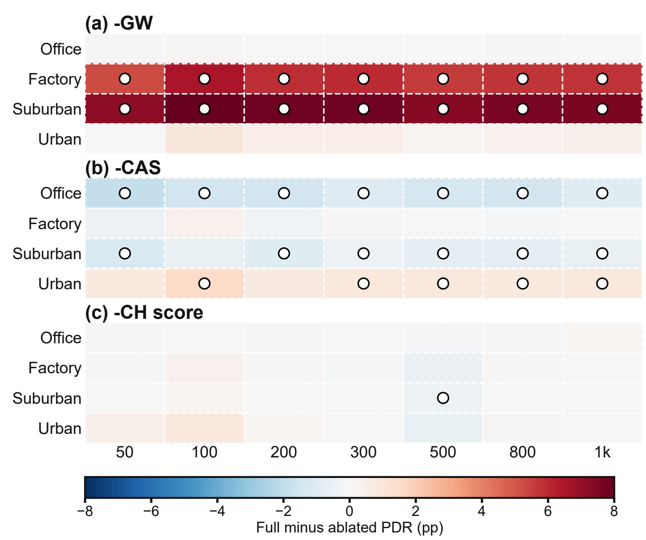

Fig. 4. Compact NS-3 AERIS ablation over the full 50–1000 node sweep ($n = 30$ per cell). Each panel is one disabled module. Color encodes full-minus-ablated PDR in percentage points; filled dots mark Holm-corrected significant cells. This single-column heatmap keeps all node scales visible without line overlap.

E. Mechanism and Trade-off

The 400-replicate-per-cell mechanism study is summarized in Fig. 5. It covers 12 environment-scale cells: office, factory, suburban, and urban at 100, 500, and 1000 nodes. Gateway uplink PDR stays near 1.0 in office, factory, and suburban cells, but drops sharply at urban-1000, exactly where end-to-end PDR collapses. A targeted rerun of the three harsh 1000-node cells on an independent machine reproduced the same bottleneck pattern to the reported precision. The visual takeaway matches the ablation: Gateway carries the gain, CAS is conditional, and Skeleton remains dormant because Gateway support resolves most eligible fallback cases in this configuration. These cell-level FND values expose harsh-regime role exhaustion and are not directly comparable to the pooled 100-node block in Table III.

The trade-off is also clear. Table III shows that AERIS records the lowest consumed energy (131.1 J), but this value reflects early forwarding-role exhaustion rather than an isolated efficiency advantage. It also has the shortest lifetime (215.5 rounds), confirming that delivery is obtained through concentrated relay burden. The PDR improvements quoted in the abstract are computed from the same fixed 100-node block pooled across the four environments. Fig. 5 localizes the cost to early node death and shrinking two-hop support in harsh cells, but its FND bars are cell-level mechanism measurements rather than the same pooled statistic used in Table III.

Figures 2–4 and Table III suggest a concrete deployment rule. Prefer PEGASIS-like routing when lifetime dominates in benign cells. Consider mature CTP/RPL-style stacks first when richer collection-tree control is acceptable, especially in office, factory, or urban regimes. Choose AERIS when delivery under heterogeneous harsh links is the priority and a simple auditable controller is required, especially in suburban-like cells. Office-like regimes, longevity-first deployments, and mature LLN stacks are the clearest negative cases.

V. Conclusion

This paper evaluates AERIS through a five-protocol NS-3 comparison, an expanded seven-protocol boundary sweep against stronger collection-tree-style baselines, a collision-aware stress layer, and a mechanism study. The claim is deliberately bounded: AERIS is a reliability-first alternative to classical WSN baselines, but richer collection-tree routing narrows or removes that advantage in office, factory, and urban regimes. The current audit uses simplified CTP/RPL-MRHOF-style baselines, an idealized canonical MAC layer, static topologies, and $\mathrm{PDR}_{\mathrm{expected}}$ for the event-triggered TEEN baseline. Within those limits, AERIS's main contribution is a visible deployment boundary, not a universal ranking. Future work should compare against full RPL/ORW/ORPL-style stacks and test whether Gateway-style reinforcement can coexist with collection-tree parent selection without reproducing the early-node-death cost.

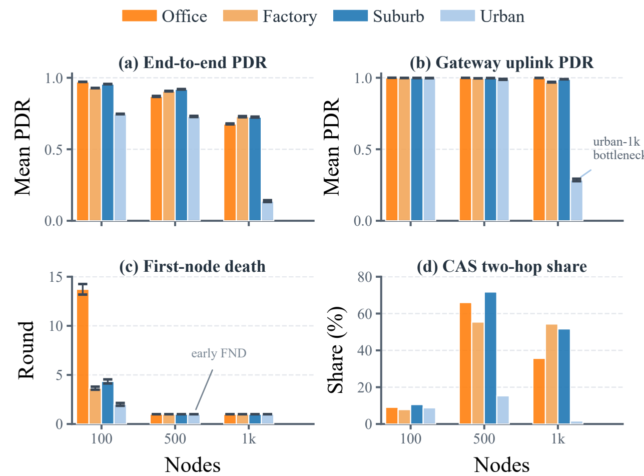

Fig. 5. AERIS mechanism study from 400 replicates per environment-node cell across 12 cells (4800 runs total). The four panels use grouped bars to show end-to-end PDR, Gateway-uplink PDR, first-node death (FND), and CAS two-hop share by environment and node scale.

TABLE III
FIXED 100-NODE POOLED TRADE-OFF SUMMARY.

Method	PDR↑	Consumed J↓	Lifetime↑	FND↑	Hops↓
LEACH	0.260	165.4	300.0	NR	1.71
HEED	0.414	165.2	300.0	11.4	2.00
TEEN	0.432	159.4	299.0	NR	1.28
PEGASIS	0.373	137.6	300.0	70.7	32.37
AERIS	**0.674**	**131.1**	**215.5**	**15.8**	**1.98**

Note: Values are pooled across the four environments. The broader sweeps in Figs. 2–4 cover 50–1000 nodes. NR = no first-node death before the 300-round horizon. Bold indicates AERIS.

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